

Nudging Academic Responsibility

Investigating social media use among Apple users through behavioral insights

Authors

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Abstract

The pervasive use of social media among university students has raised significant concerns regarding its impact on academic performance and overall well-being. This paper investigates the excessive time spent on social media platforms, particularly among Luiss University students using Apple mobile devices, and proposes a behavioral intervention aimed at reducing this phenomenon. Utilizing the COM-B framework, which emphasizes the interplay of capability, opportunity, and motivation, we establish a baseline for social media usage through a short survey. The intervention employs targeted advertisements to enhance awareness of Apple's screen time control features, encouraging students to prioritize their

academic responsibilities over social media engagement. Our findings indicate a clear preference for "rewarding" advertisements over "punishing" ones, suggesting that positive reinforcement is more effective in promoting responsible media use. By nudging students towards healthier social media habits, we aim to foster increased productivity, reduced stress, and improved academic achievement. This research highlights the urgent need for behavioral interventions that not only address the negative consequences of excessive social media use but also promote the positive potential of digital engagement when utilized for educational purposes. Future efforts will explore strategies to optimize screen time for academic success while maintaining a balanced digital lifestyle.

Keywords

Social Media, University Students, Behavioral Intervention, Screen Time Management, COM-B Framework, Behavioural Public Policies, Targeted Advertising, Digital Well-Being, Positive Reinforcement, Responsible Media Use, Nudging.

Introduction

The emergence of social media has given rise to a wide range of worries about how

it will affect people, particularly university students. University students frequently find themselves spending a substantial amount of time online, ignoring their studies and academic duties, due to the proliferation of mobile devices and social media programmes like TikTok. With an emphasis on their habit of spending too much time on social media apps on their Apple mobile phones rather than studying for exams, we will investigate the subject of controlling social media use among university students in this essay.

Our goal is to raise the proportion of students who use Apple's controlling mechanism to limit their use of social media. The COM-B paradigm, which contends that all behaviour has three parts: capability, opportunity, and motivation, will be used in this instance. Through a short survey and data collection, we will establish a baseline for social media use among students and set a goal for reducing it. We hope to use targeted advertising to increase people's awareness of and familiarity with the Apple product possibilities. The purpose of our research is to persuade college students to prioritise their academic responsibilities over other commitments and set time limits for online activity. This paper will examine the causes of excessive social media use, the value of raising awareness, and the potential results of our behavioural intervention.

This behavioural intervention is important as the negative effects of excessive social media use are well known and have the potential to significantly affect the lives of young people, especially university students who are subject to a variety of demands on their time and attention. The intervention might boost productivity, lessen stress, and enhance academic achievement by limiting the amount of time students spend on social media. Student development of better sleep hygiene, which is crucial for overall wellbeing, may also result from encouraging healthy social media usage habits. It should be noted that social media addiction is a sizable issue that can be avoided in the long run by addressing it early on, as in college. This intervention may benefit students' long-term physical and mental health by assisting them in finding a better balance between social media use and other crucial activities like studying, interacting with others, and exercising.

To effectively reach a broad audience of university students, it is practical and economical to employ targeted advertisements to raise knowledge of the alternatives for limiting social media use on Apple mobile devices. Because of the widespread use of digital media, which exposes students to ads regularly, it is a perfect medium for spreading knowledge

and encouraging positive behaviour change.

We can employ a combination of positive and negative language, as well as reward and guilt strategies, to make these customised commercials attention-grabbing. We can emphasise the advantages of reducing social media use, such as enhanced academic achievement, decreased stress levels, and improved sleep hygiene, for positive messaging. We can underline how Apple's controlling system can support students in maintaining concentration and achieving their objectives as we highlight its features. We can display the dangers and unfavourable effects of excessive social media use, such as addiction, lost productivity, and detrimental effects on mental health, in order to send a negative message.

This message can act as a warning to students who may not be aware of the risks associated with unrestricted social media use. We may reward students who use social media responsibly by including incentives in our targeted marketing. For instance, we could give students who consistently limit their social media use and succeed academically discounts on Apple products or other incentives. In order to motivate kids to behave, we can employ guilt-based strategies. We can draw attention to the negative effects overuse of social media can have on

students' mental health, academic performance, and general wellness. We can motivate students to take ownership of their behaviour and make constructive changes by presenting their usage of social media as a choice that can either benefit or hurt them. Overall, we can produce tailored advertising that are both effective and eye-catching in persuading college students to limit their social media use by mixing positive and negative messages, reward and guilt strategies, and a focus on Apple's controlling mechanism.

Literature Review

In today's digital age, social media has become integral to our daily lives. Social media platforms offer many benefits, from communicating with friends and family to staying up-to-date with the latest news and trends. However, as the use of social media continues to grow, concerns have been raised about its impact on academic performance. Numerous studies have explored this direct relationship between social media usage and academic success. In this literary review, we will examine these studies in-depth to understand better how social media use affects academic performance among university students, who use media more than anyone else. The latter has become the primary victims; this may be because social media

is an appealing way for students to avoid boredom when studying or searching for course material online, diverting their focus away from their duties. Through this literary review essay, we aim to explore the negative side of the debate surrounding the impact of social media usage on students' academic performance by analysing various scholarly sources from different perspectives.

In 1995, Robert Putnam theorised the time displacement theory to shed light on why media use has a detrimental effect on academic accomplishments.¹ This theory suggests that due to the appealing nature of media platforms, students are more inclined to devote much of their time to indulging in them rather than focusing on academic pursuits. Furthermore, some individuals may even skip classes and spend their time using media instead. In short, this perspective views the media as a tool that serves as a distraction for students and that hampers their ability to perform well scholarly by reducing their productivity and efficiency levels.

Sociologists such as Ravizza, Hambrick, and Fenn (2014) state that non-academic social media use is negatively related to classroom performance among university students². According to Oye, Adam, and Nor Zairah (2012), academic performance was negatively impacted when social networking sites were used solely for social and non-academic purposes³. In 2013, it was discovered that students tend to use media for personal use in activities such as online shopping or chatting, which can cause procrastination, time-wasting, and reduce work quality and productivity.

Furthermore, in the “model of perceived influence on academic performance using social networking”, it is stated that non-academic media use has a detrimental impact on three areas of performance: academic results, study-related behaviours, and attitudes⁴. In fact, by using their mobile phones without any educational purpose, university students not only distract themselves but also lose track of time and wait until the last minute to study. As a result of this type of

¹ Putnam, R. D. (2000). *Bowling Alone: The Collapse and Revival of American Community*. Simon & Schuster.

² Ravizza, S. M., Hambrick, D. Z., & Fenn, K. M. (2014). Non-academic internet use in the classroom is negatively related to classroom learning regardless of intellectual ability. *Computers & Education*, 78, 109-114.

³ Oye, N. D., Adam, M. H., & Nor Zairah, A. R. (2012). Model of Perceived Influence of Academic Performance Using Social Networking. *International Journal of Computers & Technology*, 2(1), 24-29.

⁴ Ibidem.

superficial study, students take lower grades⁵.

In a survey undertaken by a group of academics at the London School of Economics, the majority of students spend more time on social media, diverting their focus away from the learning process⁶. According to university professors, social media is a continual distraction in the classroom, and as a result, alums who use it in class do not focus or learn anything. Although it may be true that they still learn in a passive way, academic research has proven that students barely remember 15% of the information given during class.

In the survey conducted by Ozer (2009), it was found that more than half of their 784 participants self-reported regular academic procrastination due to excessive use of social media⁷. It was noticed that there was a difference based on gender: male students expressed significantly

more frequent procrastination than female ones. Furthermore, there have also been documented disparities in the reasons why females and males procrastinate. While females reported delaying more because they are generally afraid of failing and seeming sluggish, males, on the other hand, cited procrastination as a result of getting more easily distracted by apps like TikTok or even gambling ones⁸.

Jean Twenge's research for San Diego University has provided evidence that excessive screen time among students can lead to significant changes in their social-emotional skills and behaviour⁹. The study specifically highlights that alumni who spend more than three hours a day on their phones are at a higher risk of experiencing depression and suicidal tendencies. This alarming finding is linked to the fact that many young people struggle with communicating effectively in face-to-face encounters, which could be caused by excessive use of screens.

⁵ Kushlev, K., Proulx, J. D. E., & Dunn, E. W. (2017). Smartphone use undermines enjoyment of face-to-face social interactions. *Journal of Experimental Social Psychology*, 78, 233-239.

⁶ Davies, G. (2021, October 15). The big problem with digital smallness. LSE Business Review. Retrieved from <https://blogs.lse.ac.uk/businessreview/2021/10/15/the-big-problem-with-digital-smallness/>

⁷ Uzun Ozer, B., Demir, A., & Ferrari, J. R. (2009). Exploring Academic Procrastination Among Turkish Students: Possible Gender Differences in Prevalence and Reasons. *The Journal of Social Psychology*, 149(2), 241-257.

⁸ Booker, C.L., Kelly, Y.J. & Sacker, A. (2018) Gender differences in the associations between age trends of social media interaction and well-being among 10-15 year olds in the UK. *BMC Public Health*. <https://doi.org/10.1186/s12889-018-5220-4>

⁹ Twenge, J. (2022). The Effects of Social Media on Teens and Young Adults. Retrieved from San Diego State University Psychology Department website: <https://psychology.sdsu.edu/social-media-and-kids-mental-health-jean-twenge/>

Moreover, such individuals may also struggle with building deep and lasting relationships due to their dependence on social media, gaming, and other forms of e-interaction. For instance, excessive screen time during quarantine had shown its effects after the pandemic emergency, in which students had more difficulties communicating with each other and most of them struggled with depressive thoughts¹⁰.

According to Constance Scharff's 2010 study, the behaviours associated with excessive phone use may signify maladaptive brain responses. These findings are significant and highlight the need for various stakeholders, such as students, parents, educators, and policymakers, to pay close attention to the impact of technology on young people's lives. It is crucial that we strive to cultivate healthy peer interactions and encourage positive mental health outcomes among students by reducing screen time as much as possible¹¹. With an increase in technology usage among younger individuals, it is more important

than ever before that we collectively work towards minimising harmful side effects related to constant screen use.

According to Oxford University, the current generation of students is spending an immense amount of time on their electronic screens. As a matter of fact, the average amount of time spent comes out to around seven hours and 22 minutes per day; it is relevant to say that this statistic does not even consider the additional screen time that students spend on their computers or phones for academic purposes. This addiction to screens is having a detrimental effect on the quality of students' sleep¹². Due to excessive screen time, students are experiencing a lack of concentration during daily university activities such as attending classes and studying. Consequently, this has led to a decreased ability to learn, concentrate, and remember class material.

Many academics highlight the essentiality of monitoring screen usage carefully since it can seriously affect one's productivity and overall mental and physical well-being. Liebherr et al.

¹⁰ Houghton, S., Lawrence, D., Hunter, S. C., Rosenberg, M., Zadow, C., Wood, L., & Shilton, T. (2018). Reciprocal Relationships between Trajectories of Depressive Symptoms and Screen Media Use during Adolescence. *Journal of youth and adolescence*, 47(11), 2453–2467. <https://doi.org/10.1007/s10964-018-0901-y>

¹¹ Viner, R. M., Gireesh, A., Stiglic, N., Hudson, L. D., Goddings, A. L., Ward, J. L., & Nicholls, D. E. (2019). Roles of cyberbullying, sleep, and physical activity in mediating the effects of social media use on mental health and wellbeing among young people in England: a secondary analysis of longitudinal data. *The Lancet. Child & adolescent health*, 3(10), 685–696. [https://doi.org/10.1016/S2352-4642\(19\)30186-5](https://doi.org/10.1016/S2352-4642(19)30186-5)

¹² Hale, L., & Guan, S. (2015). Screen time and sleep among school-aged children and adolescents: a systematic literature review. *Sleep medicine reviews*, 21, 50–58. <https://doi.org/10.1016/j.smrv.2014.07.007>

(2020) suggest that the increased use of smartphones may have negative impacts on the performance of tasks required for everyday study-related activities¹³. For instance, constantly checking and responding to incoming messages on the phone while studying can interfere with course content absorption.

Moreover, engaging in enjoyable activities such as browsing through social media or playing games on a smartphone can lead to a state of flow experience, which may cause difficulty in focusing on academic work. Montag and Diefenbach (2018) explain that interruptions due to social media notifications or other distractions can break a student's flow state, resulting in decreased motivation and productivity during the task at hand¹⁴. This highlights the importance of minimising smartphone use during study time to achieve higher levels of concentration and efficiency in learning. The social media platform, TikTok, is widely recognized for enabling the creation and distribution of concise video clips. Its popularity has skyrocketed in recent years, and its extensive reach and

influence have raised concerns about its effects on various aspects of life, including academic performance¹⁵. Studies suggest that excessive usage of social media platforms like TikTok can lead to decreased concentration levels, poor time management skills, and increased procrastination. The captivating content on TikTok characterised by infinite scrolling and stimulating videos can be particularly detrimental to students attempting to concentrate on their studies. Research shows that excessive use of social media platforms results in sleep cycle disruptions and symptoms such as fatigue, reduced attention span, and impaired memory recall. The compromised quality of sleep negatively affects the ability to concentrate and retain information, which is critical for achieving academic success. One of the key problems with this form of media is that it promotes a culture where individuals are constantly exposed to carefully crafted displays, leading to social comparison. This type of content consumption often leaves students feeling inadequate or anxious, ultimately

¹³ Liebherr, M., Schubert, P., Antons, S., Montag, C., & Brand, M. (2020). Smartphones and attention, curse or blessing? - A review on the effects of smartphone usage on attention, inhibition, and working memory. *Computers in Human Behavior Reports*, 1, 100005. doi:10.1016/j.chbr.2020.100005

¹⁴ Montag, C., & Diefenbach, S. (2018). Towards Homo Digitalis: Important Research Issues for Psychology and the Neurosciences at the Dawn of the Internet of Things and the Digital Society. *Sustainability*, 10(2).

¹⁵ Lobo, J., Dimalanta, G., Bautista, C., Buan, E., & De Dios, D. A. (2022). TikTok Consumption and Level of Class Engagement of Performing Arts Students in the New Normal: Destructive or Beneficial?. *American Journal of Education and Technology*, 1(1), 1–9. <https://doi.org/10.54536/ajet.v1i1.305>

resulting in lowered self-confidence which can negatively impact their academic focus and performance.

In light of the overwhelming evidence indicating that social media can be addictive and detrimental to one's productivity, adequate measures have been taken to limit its negative effects. One of the examples of such a measure is the implementation of tools like Apple Time Limit. It has come to light that on an average day, we subconsciously check our phones at least 58 times without even realising it. This holds especially true for students pursuing higher education - they remain unaware of the alarming amount of time they spend online. Fortunately, with the introduction of Apple's time limit feature, students are becoming increasingly aware of their phone usage and are cutting back on screen time. The benefits are twofold - not only do they use their phones less often, but they also divert their focus towards academic pursuits instead of getting side-tracked by social media distractions.

In conclusion, the influence of social media on students' performance indicates that excessive use of these platforms can result in a lack of time dedicated to studying. The frequency at which social media is used directly correlates with decreased academic progress and attention span. Although social networking sites may offer learning

opportunities, constant use can lead to negative consequences such as procrastination, poor communication skills, isolation among students, lack of sleep, body dysmorphia, and depressive thoughts.

Fortunately, Apple's time limits provide an effective solution for regulating usage and better understanding the adverse effects of spending too much time on these platforms. By setting restrictions on screen time or using apps like "Freedom" or "FocusMe," students can limit access to non-academic-based social media activities during specific study periods. Based on these concerns related to the excessive use of social media platforms by students, we can hypothesize that those who engage in non-academic-based activities will likely encounter poorer academic outcomes than those who do not indulge in these activities frequently.

H1: The greater the use of social media for non-academic use, the poorer the academic performance.

Behavioural Intervention

It is an indisputable fact that social networks are now an integral part of our daily lives. Facebook, Twitter, Instagram, TikTok... these are just some of the most popular and widely used social networks that record daily shares, updates and

appreciations from all parts of the world. They have established themselves as the main means of communication in our lives in an incredibly rapid manner. Precisely because of their relevance, we selected social media and their use as the target of our behavioural intervention. Therefore, our first step was to choose this as the main policy area and next, we started working on this macro area and we went on to perform our behavioural reduction. We drew the different strategic domains and decided to focus our intervention on reducing the recreational use of social media on mobile phones made during study hours.

While we were listing some of the possible decisions and behaviours for our strategic domain we came across a feature, found in the settings of Apple's operating system that allows to manage the usage functions of all applications on the device including social media programmes. Among the features of this option, there is the possibility to enter a limitation on the time of use, as well as to monitor it. Although to this day Apple boasts more than one billion iPhone owners worldwide, most of these users are not aware of this feature. In fact, they often resort to other external applications that are downloaded to achieve the same purpose.

Despite this, the tool offered by Apple offers a wide range of functionalities,

including several incentives to reduce the amount of time spent browsing social media. Among these, we find the possibility to select the maximum number of minutes or hours we wish to spend on a given application. During our use we will receive notifications designed to update us on the time spent, until once the fixed limit is reached the application will automatically shut down.

All of these strategies could prove to be extremely effective in reducing usage time, but above all, they could spread awareness about the time that is often wasted on social rather than other more remunerative activities.

For this reason, we thought, with our behavioural intervention, we would incentivize the use of such a function, encouraging its spread and hoping to achieve positive results regarding a limitation of the time of social media use.

Continuing with behavioural mapping, once we determined that the behaviour we want to promote is the use of this function to achieve the desired outcome of reducing social media use, we asked ourselves who the recipients of our intervention should be. In particular, we decided to aim with our intervention at changing the habits of university students who often, during periods of intense study in the middle of exam sessions, waste part of the time they could be spending studying, surfing online. In fact,

using the Internet, whether to chat on social networks or to search for information and insights, does not help the learning process.

According to a study by the Department of Sociology at Milan's Bicocca University, for every hour spent studying with the internet, learning drops. Moreover, 5-10% of online users are unable to control the time spent on social media. The reason is the immediate gratification, with little effort, offered by social media, which causes the brain to develop dependence on the stimuli they offer.

To sum up, for the general issue of regulating Social Media use, we decided to focus on the behaviour of university students who spend a huge amount of time on social media applications on their apple mobile phones rather than studying for their exam session. The behaviour we wish to change is the fact that university students ignore the existence of an Apple controlling mechanism that could enable them to keep track of the amount of time they spend navigating on social media and consequently, it would help them to reduce it. Therefore, the objective that we want to achieve is to increase the number of students using the instruments provided by Apple and consequently reduce the time they spend online on social media apps. As for our baseline, not having found any kind of pre-existing

statistics that met our needs, we decided to collect data ourselves regarding the time spent on social media by university students, and in particular of our colleagues here at Luiss Guido Carli University. This was accomplished through a survey that we submitted to several students. The primary question in our survey was "How much time per day do you spend on social media?" We have predisposed four possible answers to this question:

1. Less than an hour.
2. Between one and two hours.
3. Between two and three hours.

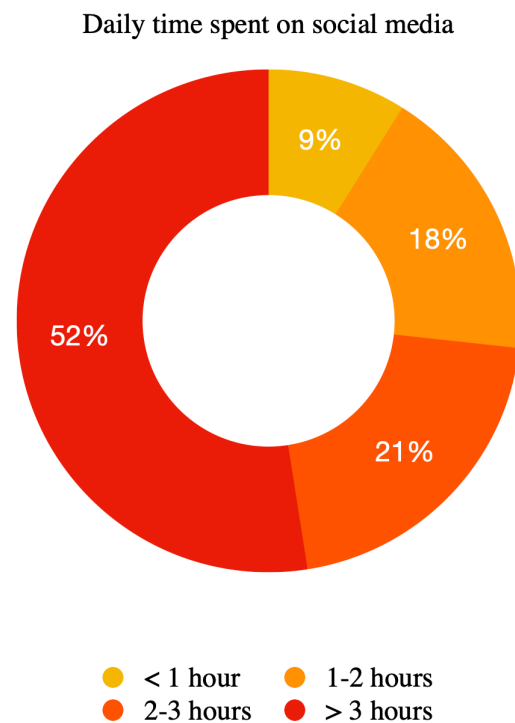


Figure 1. Survey results from 83 respondents indicate daily time spent on social media, with categories ranging from less than 1 hour to more than 3 hours.

4. More than three hours.

Analysing the 83 responses obtained from the survey (*Figure 1*), it is clear that most of our respondents use social media for more than three hours a day.

Once we collected these initial data, we were able to set a goal in terms of decreasing the time of use. Considering that the majority of our respondents (52%) answered that they use social media for more than three hours a day, we set as our main goal the reduction of the time of use, to make it fall between one and two hours.

Regarding the COM-B model, we used this framework as a systematic approach to analyse the factors that influence our targeted behaviour and to guide us in the development of our intervention to promote behaviour change. The COM-B model proposes that there are three components to any behaviour: Capability, Opportunity and Motivation. As each of these components interact, interventions must target one or more of these in order to deliver and maintain effective behaviour change.

In this case, to understand what may be the drivers of the behaviour of students, who often abuse social media, we analysed all three components and determined how the COM-B structure is characterized in this specific context. The first component, capability, refers to the individual's psychological and physical

capacity to engage in a particular behaviour. It includes both the physical ability and the knowledge or skills required to perform the behaviour. In this case, individuals' capability to perform such behaviour could be affected by the lack of knowledge: university students may ignore that there are existing app options that could help them control and limit the time they spend on social media apps. The second component, opportunity, refers to the external factors that enable or hinder the behaviour. It includes environmental and social influences that affect it. Furthermore, it can both physical and psychological and, in this instance, it is constituted by the fact that mobile phones and social media apps are easily available and using them does not require any kind of physical effort or mental effort and but in turn, it increases our levels of dopamine. In fact, scientists have shown how social media can trigger mechanisms linked to the brain centres of pleasure.

Every time we see a large number of *likes*, for example, our level of dopamine, commonly known as the happiness hormone, increases. Social networks, in short, exploit our brain's ability to generate sudden bursts of pleasure to keep us hooked and spending as much time as possible on the apps.

Finally, coming back to the COM-B model, we find motivation. It could be

described as the individual's conscious or unconscious processes that energize and direct behaviour. It can be both reflective or automatic and therefore, in this situation, the social media may be considered as an easy and accessible distraction during study breaks for university students (reflective motivation) or students may develop the habit of constantly scrolling on social media almost automatically and often lose track of time when doing it (automatic motivation).

Among all the possible behavioural entry points we have decided to focus our behavioural intervention on the component of capability. In fact, our project consists in spreading knowledge through advertisements on apple's function, to encourage university students to limit the time they spend online. We decided to prioritize such an entry point as it is both feasible and effective: it is possible and relatively easy to spread awareness and knowledge about the different options offered by Apple, in order to promote their use among young university students. This could be done through the use of targeted advertisements. Finally yet importantly, spreading information is very critical in order to correct the behaviour of university students.

Now, focusing on what will be our actual intervention, we have come up with two

samples of small advertisements that could be used for this purpose. In particular, the two videos differ in regard not so much to the message we want to communicate (social media should not hinder university studies) but especially in the way this message is conveyed. As a matter of fact, in the first video we decided to use a reward technique, while in the second we employed a language aimed at stimulating guilt in the viewer.

Obviously, the idea of using the rewarding technique depends on the fact that the concept of rewards and incentives is often used in behavioural interventions to promote behaviour change. It involves providing positive reinforcement or incentives to individuals. Rewards can be tangible (e.g., monetary incentives, gifts, or privileges) or intangible (e.g., praise, recognition, or social approval).

In our case, our intangible reward to reducing time online and rather spend it studying, would be accomplishing university studies and getting a degree. The basic premise of this mechanism is that by associating a behaviour with a positive consequence or reward, individuals are more likely to engage in and sustain that behaviour over time. Thus, rewards can help reinforce desired behaviours, increase motivation, and create a positive association with the target behaviour.

On the other hand, we chose to use guilt-inducing language for the same reason. In the realm of behaviour change interventions, guilt can be leveraged as a persuasive tool. It is an emotional response that arises when an individual believes they have violated a moral or social norm, or when they perceive themselves as responsible for causing harm or negative outcomes. Guilt can serve as a powerful motivator for behaviour change, as individuals may seek to alleviate the guilt, they feel by taking corrective actions or making amends. Finally, although we did not conduct our own evaluation, we still imagined a way in which we could test our intervention.

After the first survey regarding the number of hours spent on social media, we planned to conduct another one, again on the same sample, in order to be able to verify any eventual reduction in time of use. The idea would be to divide our sample into two groups. To the former we would submit the first version of the commercial and to the latter the second one. The final survey would be conducted one week after the advertisements were submitted, to see if they had the desired effect and nudged students to decrease the time they spent online. Plus, in addition to assessing whether the ads fulfilled their function, we could also compare them and analyse if one of them worked better than the other and which one it is.

Conclusion

To conclude, in this paper we focused on the harmful aspects of social media and went through all the relevant literature and studies that testify to the negative effects that the excessive use of these tools can cause. As social networks are so popular nowadays, above all among the younger generation, we found it of utmost importance to act in this area and create our own behavioural intervention with the aim of influencing young people to regulate their use of social networks.

We thought of a simple and accessible way to convey this message, devising short videos that could promote the use of control and limitation options available on all iPhones and Apple devices. We have constructed and elaborated the structure of these ads based on the notions provided by behavioural sciences and using evidence-based techniques, with the aim of appealing to the emotions of young university students.

However, we could not test the effectiveness of both videos over time, therefore we were not able to test whether our ultimate goal, to reduce social media usage time to a daily average of between one and two hours, can be achieved through this intervention.

Nevertheless we decided to conduct another survey, in order to understand whether in the immediate term these commercials have had the desired effect

and pushed university students to think about using media more responsibly through the tools provided by Apple. Luiss University students, after seeing both options, were asked to indicate whether they preferred the “rewarding” advertisement or the “punishing” one and which of them had had the greatest impact on them. A large majority stated the rewarding one resulted more effective and fun to watch. They believed the message of the punishing advertisement was too exaggerated.

However, both videos had a specific function within our research. The rewarding one played with the idea of a clock constantly ticking to remind students they must begin studying when they reached the threshold of screen time, they were allowed to that day. The punishing advertisement is focused on generating a sense of guilt for wasting time on social media, rather than focusing on one’s own exams. It displays a pessimistic perspective underlining all the possible terrible consequences of not limiting the amount of time spent online.

The main goal we wanted to achieve through the videos was indeed nudging people in adopting a specific behaviour, without making it look like an obligation. By uploading the advertisements on social media, we wanted to raise awareness on the existence and utility of

Screentime, a function provided by Apple, which is oftentimes overlooked.

Time spent on social media by younger generations is increasing over time. This increasing trend is extremely worrying when looking at university students. When studying for their exams, rather than taking a break and breathe some fresh air, they prefer to scroll on their favourite social media apps.

It was found that 97% of them uses social media application and only a 1% employs them for academic purposes. Such results underline how social media can become a form of distraction for university students not only affecting their academic performance but also social interactions, sleep duration and encouraging a sedentary life.

College students tend to use online platforms more frequently than other social categories, causing them unrepairable damage. They often suffer from burnouts, so have a hard time disconnecting from the online world. In worse cases, students could also be subject to severe mental health declining.

For this reason, once again knowing how much time they spend on social media can generate a sense of guilt and persuade them to set a limit to the amount of time spent on it. After being encouraged to engage in alternative activities they could finally reconnect with their own realities.

However, what is also true is that “All screen use is not equal” (Michal Robb). Nowadays we have access to almost unlimited information and tend to use this opportunity the wrong way. According to this last statement, it was proven that screen time can also be useful and have a positive influence in supporting education. This is why the amount of time spent surfing the internet should be employed more wisely, for academic purposes. So, a further step after promoting of App Limits on iPhones, could be encouraging students to use their limited online time for didactic purposes.

Bibliography

- Booker, C. L., Kelly, Y. J., & Sacker, A. (2018). Gender differences in the associations between age trends of social media interaction and well-being among students in the UK. *BMC Public Health*.
- Davies, G. (2021, October 15). The big problem with digital smallness. *LSE Business Review*. Retrieved from <https://blogs.lse.ac.uk/businessreview/2021/10/15/the-big-problem-with-digital-smallness/>
- Hale, L., & Guan, S. (2015). Screen time and sleep among university students: A review. *Sleep Medicine Reviews*, 21, 50–58. <https://doi.org/10.1016/j.smrv.2014.07.007>
- Houghton, S., Lawrence, D., Hunter, S. C., et al. (2018). Reciprocal relationships between trajectories of depressive symptoms and screen media use in young people. *Journal of Youth and Adolescence*.
- Kushlev, K., Proulx, J. D. E., & Dunn, E. W. (2017). Smartphone use undermines enjoyment of face-to-face social interactions. *Journal of Experimental Social Psychology*, 78, 233–239.
- Liebherr, M., Schubert, P., Antons, S., Montag, C., & Brand, M. (2020). Smartphones and attention, curse or blessing? - A review on the effects of smartphone usage on attention, inhibition, and working memory. *Computers in Human Behavior Reports*, 1, 100005. doi:10.1016/j.chbr.2020.100005
- Lobo, J., Dimalanta, G., Bautista, C., Buan, E., & De Dios, D. A. (2022). TikTok Consumption and Level of Class Engagement of Performing Arts Students in the New Normal: Destructive or Beneficial?. *American Journal of Education and Technology*, 1(1), 1–9. <https://doi.org/10.54536/ajet.v1i1.305>
- Montag, C., & Diefenbach, S. (2018). Towards Homo Digitalis: Important research issues for psychology and the neurosciences at the dawn of the internet of things and the digital society. *Sustainability*, 10(2).
- Oye, N. D., Adam, M. H., & Nor Zairah Ab. Rahim. (2012). Model of perceived influence of academic performance using social networking. *International Journal of Computers & Technology*, 2(1), 24–29.

Putnam, R. D. (2000). *Bowling alone: The collapse and revival of American community*. Simon & Schuster.

Ravizza, S. M., Hambrick, D. Z., & Fenn, K. M. (2014). Non-academic internet use in the classroom is negatively related to classroom learning regardless of intellectual ability. *Computers & Education*, 78, 109–114.

Twenge, J. (2022). The Effects of Social Media on Teens and Young Adults. Retrieved from San Diego State University Psychology Department website: <https://psychology.sdsu.edu/social-media-and-kids-mental-health-jean-twenge/>

Uzun Ozer, B., Demir, A., & Ferrari, J. R. (2009). Exploring academic procrastination among Turkish students: Possible gender differences in prevalence and reasons. *The Journal of Social Psychology*, 149(2), 241–257.

Viner, R. M., Gireesh, A., Stiglic, N., Hudson, L. D., Goddings, A. L., Ward, J. L., & Nicholls, D. E. (2019). Roles of cyberbullying, sleep, and physical activity in mediating the effects of social media use on mental health and wellbeing among young people in England: a secondary analysis of longitudinal data. *The Lancet. Child & adolescent health*, 3(10), 685–696. [https://doi.org/10.1016/S2352-4642\(19\)30186-5](https://doi.org/10.1016/S2352-4642(19)30186-5)